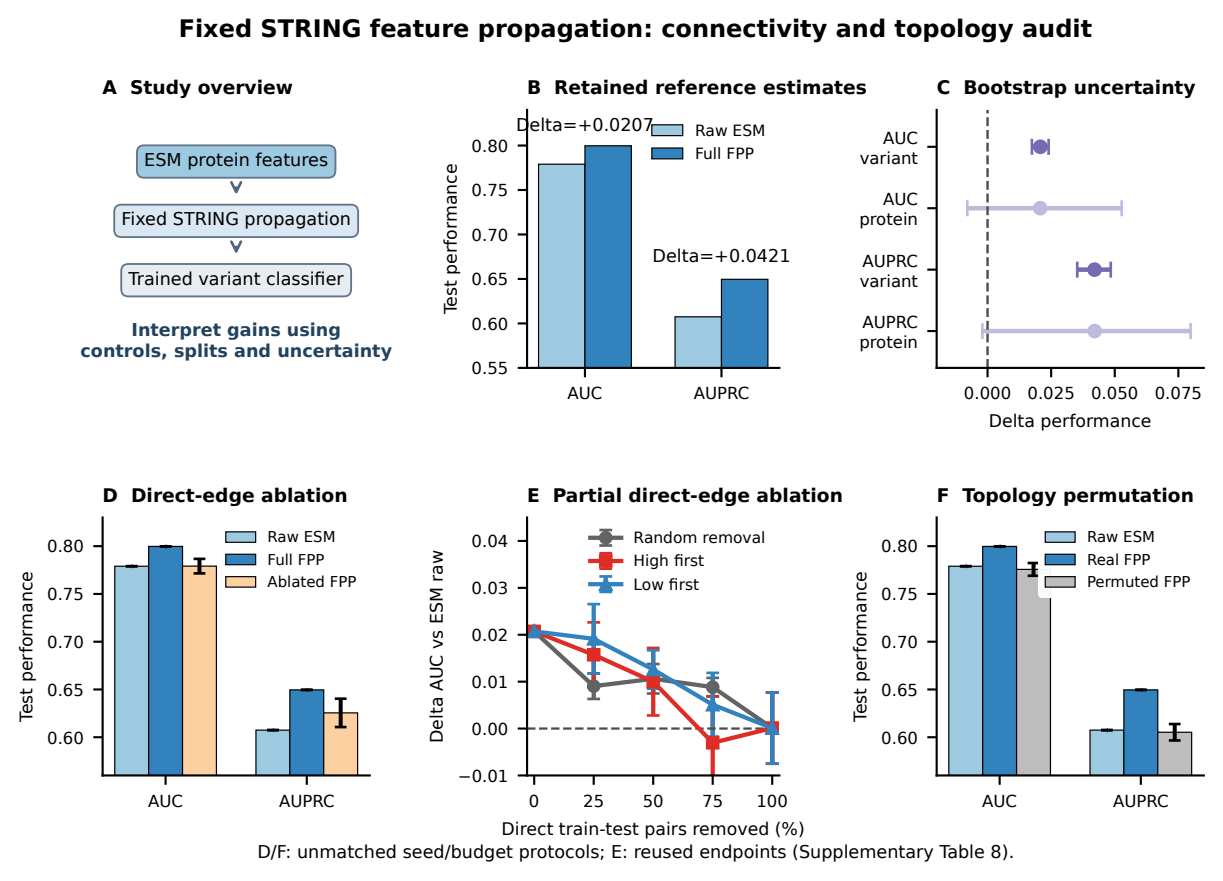
Fixed STRING feature propagation: connectivity and topology audit
A Study overview
ESM protein features
Fixed STRING propagation
Trained variant classifier
Interpret gains using controls, splits and uncertainty
B Retained reference estimates
Delta=+0.0207
Raw ESM
Full FPP
Delta=+0.0421
Test performance
AUC
AUPRC
C Bootstrap uncertainty
AUC variant
AUC protein
AUPRC variant
AUPRC protein
Delta performance
D Direct-edge ablation
Raw ESM
Full FPP
Ablated FPP
Test performance
AUC
AUPRC
E Partial direct-edge ablation
Random removal
High first
Low first
Delta AUC vs ESM raw
Direct train-test pairs removed (%)
F Topology permutation
Raw ESM
Real FPP
Permuted FPP
Test performance
AUC
AUPRC
D/F: unmatched seed/budget protocols; E: reused endpoints (Supplementary Table 8).